



Evaluation of Madras Diabetes Research Foundation-Indian Diabetes Risk Score in detecting undiagnosed diabetes in the Indian population: Results from the Indian Council of Medical Research-INDia DIABetes population-based study (INDIAB-15)

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Background & objectives: Screening of individuals for early detection and identification of undiagnosed diabetes can help in reducing the burden of diabetic complications. This study aimed to evaluate the performance of Madras Diabetes Research Foundation (MDRF)-Indian Diabetes Risk Score (IDRS) to screen for undiagnosed type 2 diabetes in a large representative population in India.

Methods: Data were acquired from the Indian Council of Medical Research-INDia DIABetes (ICMR-INDIAB) study, a large national survey that included both urban and rural populations from 30 states/union territories in India. Stratified multistage design was followed to obtain a sample of 113,043 individuals (94.2% response rate). MDRF-IDRS used four simple parameters, viz. age, waist circumference, family history of diabetes and physical activity to detect undiagnosed diabetes. Receiver operating characteristic (ROC) with area under the curve (AUC) was used to assess the performance of MDRF-IDRS.

Results: We identified that 32.4, 52.7 and 14.9 per cent of the general population were under high-, moderate- and low-risk category of diabetes. Among the newly diagnosed individuals with diabetes [diagnosed by oral glucose tolerance test (OGTT)], 60.2, 35.9 and 3.9 per cent were identified under

high-, moderate- and low-risk categories of IDRS. The ROC-AUC for the identification of diabetes was 0.697 (95% confidence interval: 0.684-0.709) for urban population and 0.694 (0.684-0.704) for rural, as well as 0.693 (0.682-0.705) for males and 0.707 (0.697-0.718) for females. MDRF-IDRS performed well when the population were sub-categorized by state or by regions.

Interpretation & conclusions: Performance of MDRF-IDRS is evaluated across the nation and is found to be suitable for easy and effective screening of diabetes in Asian Indians.

Key words Asian Indians - high-risk group - Indian Diabetes Risk Score - screening - type 2 diabetes - undiagnosed diabetes

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Introduction

Diabetes mellitus affects more than 463 million people worldwide, with an estimated 4.2 million people dying each year from it due to complications¹. While India accounts for one sixth of the world's population with diabetes (77 million)¹, with considerable occurrence at younger age (45-64 yr), a significant proportion of the diabetic population experience complications at their productive years with serious economic and social burden. Early identification of individuals with undiagnosed diabetes is therefore critical in reducing the burden of diabetic complications. Several intervention studies have unequivocally demonstrated that type 2 diabetes can be effectively prevented by lifestyle changes in high-risk individuals^{2,3}. The major task for public health administrations is to identify these high-risk individuals who would benefit from intensive lifestyle modification. However, 'universal screening', *i.e.* screening the entire population, is cumbersome and thus a cost-effective tool to selectively screen the high-risk groups would be necessary.

Development of the Indian Diabetes Risk Score (IDRS) developed by Madras Diabetes Research Foundation (MDRF), was a simple tool to aid early detection of undiagnosed diabetes in the community⁴. The MDRF-IDRS has also been shown to have additional multiple applications⁵, including predicting incident diabetes⁶, metabolic syndrome⁷, coronary artery disease⁸, non-alcoholic fatty liver disease⁹, sleep disorders¹⁰ and diabetic complications, such as peripheral vascular disease and neuropathy¹¹, and help distinguish type 2 from non-type 2 diabetes¹². The MDRF-IDRS has also been validated in several other populations¹³⁻²⁵. However, it has not been validated in a large representative population in India. The Indian Council of Medical Research (ICMR)-India DIABetes (INDIAB) study provided an opportunity to evaluate the risk score in a large representative population and thereby help detect undiagnosed type 2 diabetes in the community.

Material & Methods

The ICMR-INDIAB study is a cross-sectional, population-based survey of adults aged 20 yr and above^{26,27}. The methodological details of the study have been published elsewhere²⁶. In brief, the study sampled urban and rural residents of all the 30 states/union territories (UTs) of the country such that the total estimated sample of 120,000 individuals was representative of the entire population. The study was conducted in a phased manner. In Phase I, four regions representing the south (Tamil Nadu), north (Chandigarh), east (Jharkhand) and west (Maharashtra) of the country were studied during 2008 to 2010. Between 2011 and 2020, the remaining states were surveyed as follows; Phase II covered Andhra Pradesh (undivided), Bihar, Gujarat, Karnataka and Punjab (survey period, 2012-2013), Phase III included Delhi, Madhya Pradesh, Rajasthan and Uttar Pradesh (survey period, 2017-2018), Phase IV included Kerala, Goa, Puducherry, Haryana and Chhattisgarh (survey period, 2018-2019), North East Phase included Assam, Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim and Tripura (survey period, 2011-2017) and Phase V included Himachal Pradesh, Uttarakhand, Odisha and West Bengal (survey period, 2019-2020).

A stratified multistage sampling design was followed²⁶, wherein a three-level stratification based on geography, population size, and socio-economic status of each state was done to obtain a truly representative sample of the population from different strata. The primary sampling units were villages in rural areas and census enumeration blocks in urban areas. The ultimate stage units were households in both areas. From each household, one individual was randomly selected using the World Health Organization KISH method²⁸, thereby avoiding selection bias with respect to gender and age.

The calculated sample size of 4000 people (2800 rural + 1200 urban) provided 80 per cent power

and a five per cent alpha error in each state, assuming a type 2 diabetes prevalence of 10 per cent in urban areas and 4 per cent in rural areas, with a 20 per cent non-response rate²⁶. As a result, the overall sample size for the 30 states/UT was 120,000 people (4000×30); 113,043 individuals responded (94.2% response rate).

The analysis was conducted after excluding individuals with self-reported diabetes) on the pooled data of all the States/UTs (n=98,454), which included 28,150 urban residents and 70,304 rural residents. For the sub-analysis, the States/UTs were divided into six geographical regions as follows: North: Chandigarh, Delhi, Haryana, Himachal Pradesh, Punjab and Rajasthan; South: Andhra Pradesh (undivided), Karnataka, Kerala, Puducherry and Tamil Nadu; East: Bihar, Jharkhand, Odisha and West Bengal; West: Goa, Gujarat and Maharashtra; Central: Chhattisgarh, Madhya Pradesh, Uttar Pradesh and Uttaranchal and the North East: Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim and Tripura.

A standardized, structured questionnaire was used to collect information on demographic and socio-economic characteristics, physical activity, smoking, alcohol, medical history and family history of diabetes. Fasting capillary blood glucose was determined using a glucose meter (One Touch Ultra, Lifescan Johnson & Johnson, Milpitas, California, USA) after ensuring at least eight hours of overnight fasting. In subjects with self-reported diabetes, only the fasting glucose was measured and was excluded from the current analysis. Equipment with same specifications was used throughout the study as a measure of quality assurance.

The study and the respective sites was approved by the Institutional Ethics Committee of the Madras Diabetes Research Foundation (MDRF), Chennai, Tamil Nadu, India, and written informed consent was obtained from all the participants.

Definitions: Self-reported/known diabetes was defined by a physician diagnosis of type 2 diabetes and current use of medications for diabetes (insulin or oral hypoglycaemic agents) and were excluded from the analysis. The MDRF-IDRS was developed using multiple logistic regression model based on four simple parameters, namely age, abdominal obesity, family history of diabetes and physical activity as described elsewhere⁴. The information for these risk factors was obtained based on four questions (Table I) and one anthropometric measurement, namely the waist

Table I. Madras Diabetes Research Foundation-Indian Diabetes Risk Score (MDRF-IDRS)

MDRF-IDRS	
Particulars	Score
Age (yr)	
<35	0
35-49	20
≥50	30
Waist circumference	
Waist <80 cm (female), <90 cm (male)	0
Waist ≥80-89 cm (female), ≥90-99 cm (male)	10
Waist ≥90 cm (female), ≥100 cm (male)	20
Physical activity	
Vigorous exercise (regular) or strenuous (manual) work at home/work	0
Moderate exercise (regular) or moderate physical activity at home/work	10
Mild exercise (regular) or mild physical activity at home/work	20
No exercise and sedentary activities at home/work	30
Family history of diabetes	
No diabetes in parents	0
One parent is diabetic	10
Both parents are diabetic	20
Total score	100
IDRS score: ≥60 denotes high risk of having diabetes; 30-50 denotes moderate risk and <30 denotes that the risk of having diabetes is probably low	

circumference. Subjects with an MDRF-IDRS value of <30 was categorized as low risk, those between 30 and 50 as moderate risk and those with ≥60 as high risk for diabetes.

Statistical analysis: All statistical analysis was performed using SAS Statistical Package (version 9.4; SAS Institute, Inc., Cary, NC, USA). Estimates were expressed as mean±standard deviation or proportions. Linear regression and Cochran trend test were used to find the trend across IDRS groups for continuous and categorical variables, respectively. Student's t test was used to compare continuous variables and χ^2 tests for differences in proportions. Receiver operating characteristic (ROC) with area under the curve (AUC) was used to assess the performance of MDRF-IDRS to diagnose diabetes; $P<0.05$ was considered significant.

Results

Table II presents the characteristics of the study population based on the MDRF-IDRS risk scoring as low, moderate and high for developing type 2 diabetes. The results show that females were more likely to have high-risk score for diabetes based on MDRF-IDRS than males, in both urban and rural areas. Individuals with high MDRF-IDRS were older, had significantly higher waist circumference, higher systolic and diastolic blood pressure, were less formally educated, had stronger family history of diabetes, were less physically active and had fewer proportion of smokers and alcohol users. A similar trend was seen when stratified in urban and rural areas.

Table IIIa shows the MDRF-IDRS risk stratification by different regions of the nation. In the overall population, 14.9 per cent were in the low-risk category of MDRF-IDRS, 52.7 per cent in the moderate-risk category and 32.4 per cent in the high-risk category. It varied by regions, with the southern (38.9%) and the northern (38.3%) region having highest proportion of individuals with high MDRF-IDRS and eastern (25.4%) region having the lowest proportion of individuals with high MDRF-IDRS. A similar trend was observed when the regions were stratified as urban and rural areas (Table IIIb).

The ROC curve showing the performance of MDRF-IDRS in diagnosing diabetes among the urban and rural population and by gender is presented in Figures 1 and 2. The ROC-AUC of the MDRF-IDRS for identification of diabetes was 0.697 (95% confidence interval: 0.684-0.709) for urban and 0.694 (0.684-0.704) for rural, as well as 0.693 (0.682-0.705) for males and 0.707 (0.697-0.718) for females.

An attempt was made to determine how many of the newly diagnosed diabetic subjects found in the study using OGTT were correctly identified by MDRF-IDRS (Table IV). Of all the newly diagnosed individuals with diabetes (by OGTT), 60.2 per cent were identified as having high risk, 35.9 per cent as having moderate risk and 3.9 per cent as having low risk by the MDRF-IDRS. Over 96 per cent of all newly diagnosed diabetic subjects were identified as having either moderate or high risk by MDRF-IDRS. The values ranged from 92.1 per cent in the central region to 97.8 per cent in the southern region. In the overall population, MDRF-IDRS has a sensitivity of 60.2 per cent, specificity of 68.8 per cent, positive predictive value of 8.1 per cent, negative

predictive value of 97.4 per cent and accuracy of 68.5 per cent in identifying diabetes. A similar analysis was performed among individuals with self-reported diabetes, where 98.9 per cent of them were categorized by MDRF-IDRS as moderate/high-risk group (Supplementary Table I). Of the individuals with pre-diabetes (diagnosed by OGTT), 45.6 per cent were identified as having high risk, 45.7 per cent as having moderate risk and 8.7 per cent as having low risk for pre-diabetes by MDRF-IDRS. Overall, 91.3 per cent of all individuals with pre-diabetes were identified as having either moderate/high risk by MDRF-IDRS (Supplementary Table II).

Supplementary Table III presents the proportion of newly diagnosed diabetic subjects identified by MDRF-IDRS in the study population by individual states. MDRF-IDRS performed well in all states and did extremely well in the state of Goa where 100 per cent of the individuals, diagnosed as diabetes by OGTT, were categorized as either at high or moderate risk by MDRF-IDRS. MDRF-IDRS performed least in the state of Jharkhand where it identified 85.4 per cent of the cases as high/moderate risk. When stratified as urban or rural areas, MDRF-IDRS identifies 100 per cent of all subjects with diabetes as high/moderate risk in many urban areas of the states, including Andhra Pradesh (undivided), Assam, Goa, Haryana, Manipur, Mizoram, Nagaland, Odisha, Punjab, Sikkim and Tripura, and a few rural areas of the states, such as Goa and Rajasthan.

Discussion

The key findings of the study include the following: (i) based on the MDRF-IDRS, 32.4 per cent of the general population in India falls under high-risk category of developing type 2 diabetes; (ii) of all the newly diagnosed individuals with diabetes (diagnosed by OGTT), MDRF-IDRS identified 96.1 per cent of the population as having high/moderate risk and 3.9 per cent as having low risk for diabetes; (iii) MDRF-IDRS also performed well in urban and rural strata and also in state/ regions.

Diabetes risk scores are risk assessment tools use a combination of demographic and clinical information. Although globally several such risk scores exist²⁹⁻³⁸, their applicability to other populations is uncertain in different ethnic groups and sociocultural settings. The IDRS developed by MDRF⁴ used routinely collected, minimal information making the risk score simple, fast and more importantly, non-invasive. The variables

Table II. Characteristics of the study population stratified based on the MDRF-IDRS risk scoring^a (30 states pooled)

Variables	Urban				Rural			
	Low IDRS score	Moderate IDRS score	High IDRS score	P for trend	Low IDRS score	Moderate IDRS score	High IDRS score	P for trend
n	2750	14,262	11,138		11,897	37,621	20,786	
Male, n (%)	1409 (51.2)	6744 (47.3)	4603 (41.3)	<0.001	6807 (57.2)	18,517 (49.2)	8225 (39.6)	<0.001
Age (yr)	28.4±5.5	36.2±11.3	52.5±12.5	<0.001	29.5±6.5	40±12.8	55.5±12.7	<0.001
BMI (kg/m ²)	21.7±3.6	22.8±4.2	25.5±4.9	<0.001	20.9±3.1	21.7±3.9	23.6±4.8	<0.001
Waist circumference (cm)								
Male	78.9±9.1	82.1±10.3	90.9±12.2	<0.001	76.4±8.6	79.9±10.8	86±13.4	<0.001
Female	73.8±8.6	78.5±11.3	88.6±11.8	<0.001	71.5±8.5	75.2±11.1	84.3±12.9	<0.001
Systolic blood pressure (mmHg)	121.4±14.4	124.8±17.5	134.9±21	<0.001	122.9±13.9	126.2±17.1	134.8±21.1	<0.001
Diastolic blood pressure (mmHg)	77.9±10.7	80.1±11.1	84±11.8	<0.001	77.7±9.9	80.1±10.6	83.3±11.4	<0.001
Current smoking, n (%)	362 (13.2)	1749 (12.3)	1065 (9.6)	<0.001	1884 (15.8)	5977 (15.9)	2814 (13.5)	<0.001
Current alcohol users, n (%)	431 (15.7)	1894 (13.3)	1060 (9.5)	<0.001	2129 (17.9)	5507 (14.6)	2074 (10.0)	<0.001
Education, n (%)								
No formal schooling	367 (13.4)	1944 (13.6)	2162 (19.4)	<0.001	2733 (23)	11,018 (29.3)	8279 (39.9)	<0.001
Primary/higher secondary school	1912 (69.6)	9370 (65.8)	6808 (61.2)	<0.001	8352 (70.3)	23,303 (62.0)	11,142 (53.6)	<0.001
Technical/UG/PG education	467 (17.0)	2936 (20.6)	2152 (19.3)	0.476	799 (6.7)	3264 (8.7)	1348 (6.5)	0.004
Physical activity, n (%)								
Vigorous	557 (20.3)	356 (2.5)	5 (0.04)	<0.001	4034 (33.9)	2495 (6.6)	8 (0.04)	<0.001
Moderate	2193 (79.7)	4461 (31.3)	639 (5.7)	<0.001	7863 (66.1)	15,900 (42.3)	974 (4.7)	<0.001
Sedentary	0 (0)	9445 (66.2)	10,494 (94.2)	<0.001	0 (0)	19,226 (51.1)	19,804 (95.3)	<0.001
Family history of diabetes, n (%)								
One parent	215 (7.8)	1341 (9.4)	1909 (17.2)	<0.001	417 (3.5)	1783 (4.7)	2019 (9.7)	<0.001
Both parent	2 (0.1)	112 (0.8)	320 (2.9)	<0.001	12 (0.1)	109 (0.3)	287 (1.4)	<0.001

Values are presented as number of subjects (percentage) or mean±SD as appropriate. ^aMDRF-IDRS risk score: <30 denotes that the risk of having diabetes is probably low; 30-50 denotes moderate risk and ≥60 denotes high risk of having diabetes. SD, standard deviation; BMI, body mass index

Table IIIa. MDRF-IDRS^a in the study population by regions

MDRF-IDRS classification	North	South	East	West	Central	North East	Overall
n	22,950	15,047	11,197	13,577	7011	28,672	98,454
Low risk, n(%)	2572 (11.2)	1923 (12.8)	2104 (18.8)	1688 (12.4)	1329 (19.0)	5031 (17.5)	14,647 (14.9)
Moderate risk, n(%)	11,590 (50.5)	7277 (48.4)	6246 (55.8)	7081 (52.2)	3780 (53.9)	15,909 (55.5)	51,883 (52.7)
High risk, n(%)	8788 (38.3)	5847 (38.9)	2847 (25.4)	4808 (35.4)	1902 (27.1)	7732 (27.0)	31,924 (32.4)
^a MDRF-IDRS risk score: <30 denotes that the risk of having diabetes is probably low; 30-50 denotes moderate risk and ≥60 denotes high risk of having diabetes. North: Chandigarh, Delhi, Haryana, Himachal Pradesh, Punjab, Uttarakhand and Uttar Pradesh; South: Andhra Pradesh (undivided), Karnataka, Kerala, Puducherry and Tamil Nadu; East, Bihar, Orissa, Jharkhand and West Bengal; West: Rajasthan, Gujarat, Goa and Maharashtra; Central: Madhya Pradesh and Chhattisgarh; North East: Assam, Sikkim, Nagaland, Meghalaya, Manipur, Mizoram, Tripura and Arunachal Pradesh							

included in MDRF-IDRS are age, waist circumference, physical activity levels and family history of diabetes. Through this approach, targeted screening with non-invasive test is done for the identification of high risk groups, followed by glucose testing only in high risk individuals.

Although several risk scores available, there is no consensus on the gold-standard tools that should be used in practice. Each of these risk scores uses varying combinations of risk factors. Diabetes risk scores have also been used to predict diabetes in various populations, such as Finnish Diabetes Risk Score (FINDRISC), Finland²⁹, Danish Diabetes Risk Score, Denmark³⁰, Data from the Epidemiological Study on the Insulin Resistance Syndrome, France³¹, American Atherosclerosis Risk in Communities (ARIC), four US communities³² and Cambridge Risk Score (CRS), England³³ among others^{34,35}. The sensitivity and specificity of these scores varied widely: FINDRISC²⁹ (78% sensitivity and 77% specificity in the 1987 cohort and 81% sensitivity and 76% specificity in the 1992 cohort), DANISH³⁰ (76% sensitivity and 72% specificity), ARIC³² (52% sensitivity and 86% specificity) and CRS³³ (77% sensitivity and 72% specificity). The variables included in the risk scores are age, sex, ethnicity, body mass index (BMI), waist circumference, history of antihypertensive drug treatment, use of corticosteroids, hypertension, physical activity, family history of diabetes, smoking status, daily consumption of fruits, berries or vegetables²⁹⁻³⁵.

Risk scores developed in one population may not perform well when applied to other population. There are a few other diabetes risk scores specific for Indian population. The diabetes risk score by Ramachandran *et al*³⁶ included age, BMI, waist circumference, physical activity levels and history of diabetes

and yielded sensitivity of around 70 per cent and specificity of around 60 per cent in three tested cohorts in Chennai, Tamil Nadu. Chaturvedi *et al*³⁷ included age, waist circumference, blood pressure and family history of diabetes, which had 73 per cent sensitivity but only 56 per cent specificity. Oommen *et al*³⁸ included age, waist circumference and family history of diabetes and reported 59.5 per cent sensitivity and 60.5 per cent specificity. The applicability of the risk score to a different population is uncertain given the differences in risk factor distribution, such as varying degrees of obesity, lifestyle factors and other cultural differences. As cited above, all the risk scores had similar sensitivity and specificity and used similar clinical characteristics.

The applicability of MDRF-IDRS has also been extensively validated in several Indian populations¹³⁻²⁵ (Supplementary Table IV). Some studies^{16,21} reported that the MDRF-IDRS had good predictive value for detecting undiagnosed diabetes in the community and also suggested alternative score for MDRF-IDRS, with better sensitivity and specificity. Some studies^{14,15,17-20,25} used MDRF-IDRS in classifying the study population at risk as low, moderate and high. Despite such studies validating the applicability of MDRF-IDRS, a number of issues limit their utility, such as small sample size, lack of representative sample questioning the generalizability of the results to the larger population, differences in study design/ methodology used, lack of external validation and referral bias. The ICMR-INDIAB study, by using standardized protocol and uniform data collection procedures, provided precise details on the variation in the sensitivity in different regions of India – with the highest sensitivity being recorded in central India (67%) and lowest in eastern region (51%). Apart from

Table IIIb. MDRF-IDRS^a in the study population by regions - urban versus rural

MDRF-IDRS classification	Urban						Rural							
	North (n=6567)	South (n=4290)	East (n=3151)	West (n=3990)	Central (n=1948)	North East (n=8204)	Total (n=28,150)	North (n=16,383)	South (n=10,757)	East (n=8046)	West (n=9587)	Central (n=5063)	North East (n=20,468)	Total (n=70,304)
Low risk	517 (7.9)	355 (8.3)	450 (14.3)	343 (8.6)	221 (11.3)	864 (10.5)	2750 (9.8)	2055 (12.5)	1568 (14.6)	1654 (20.6)	1345 (14.0)	1108 (21.9)	4167 (20.4)	11,897 (16.9)
Moderate risk	3149 (48.0)	2023 (47.2)	1585 (50.3)	2013 (50.5)	1004 (51.5)	4488 (54.7)	14,262 (50.7)	8441 (51.5)	5254 (48.8)	4661 (57.9)	5068 (52.9)	2776 (54.8)	11,421 (55.8)	37,621 (53.5)
High risk	2901 (44.2)	1912 (44.6)	1116 (35.4)	1634 (41.0)	723 (37.1)	2852 (34.8)	11,138 (39.6)	5887 (35.9)	3935 (36.6)	1731 (21.5)	3174 (33.1)	1179 (23.3)	4880 (23.8)	20,786 (29.6)
Values are presented as n (%). ^a MDRF-IDRS risk score: <30 denotes that the risk of having diabetes is probably low; 30-50 denotes moderate risk and ≥60 denotes high risk of having diabetes. North: Chandigarh, Delhi, Haryana, Himachal Pradesh, Punjab, Uttarakhand and Uttar Pradesh; South: Andhra Pradesh (undivided), Karnataka, Kerala, Puducherry and Tamil Nadu; East, Bihar, Orissa, Jharkhand and West Bengal; West: Rajasthan, Gujarat, Goa and Maharashtra; Central: Madhya Pradesh and Chhattisgarh; North East: Assam, Sikkim, Nagaland, Meghalaya, Manipur, Mizoram, Tripura and Arunachal Pradesh														

the differences in the characteristics of the population studied in the different regions/states, the modest performance of the MDRF-IDRS risk score in the INDIAB population could be attributed to the data quality on the variable, 'family history of diabetes'. Although it is known that the knowledge of family history of diabetes may influence lifestyle behaviours of the individual, the reliability of the family history data is unclear. The level of awareness of individual's family history is unknown in our population, which is one of the limitations of this study.

Modifications to the risk variables included in the MDRF-IDRS were also tested. In a recent study by Venkatrao *et al*²³, the MDRF-IDRS score was tested by replacing the waist circumference component of the score with both BMI and a composite of BMI and waist circumference using data from a randomized cluster sample survey including 7496 adults at high risk for type 2 diabetes. It was reported that MDRF-IDRS using waist circumference, BMI and both BMI and waist circumference had high sensitivity (87, 88 and 82%, respectively) in detecting diabetes and use of both BMI and waist circumference in MDRF-IDRS improved its specificity. In the ICMR-INDIAB study and earlier investigations on MDRF-IDRS, waist circumference was preferred over BMI, as Indian population express Asian Indian phenotype, including increased insulin resistance and greater abdominal adiposity, *i.e.* increased waist circumference despite having a lower BMI compared to other ethnic groups.

In a country like India with a population of 1.4 billion, where more than 50 per cent of the people with type 2 diabetes remain undiagnosed, if no risk score is used, we have to screen the entire population. The logistics of such an exercise and cost can well be imagined. Hence, adding a preliminary step of using a screening score to identify at-risk individuals would have the potential to make such programs more cost-effective.

Our study has several strengths; it included states and union territories in India, and a represented both urban and rural areas. Moreover, a representative sampling frame and robust methodology were used utilizing OGTT for identification of diabetes in a large sample of 113,043 individuals with a response rate of 94.2 per cent. The high-level quality control measures implemented in the field survey ensured data reliability, allowing results to be generalized at the country level. However, there were also some limitations to our

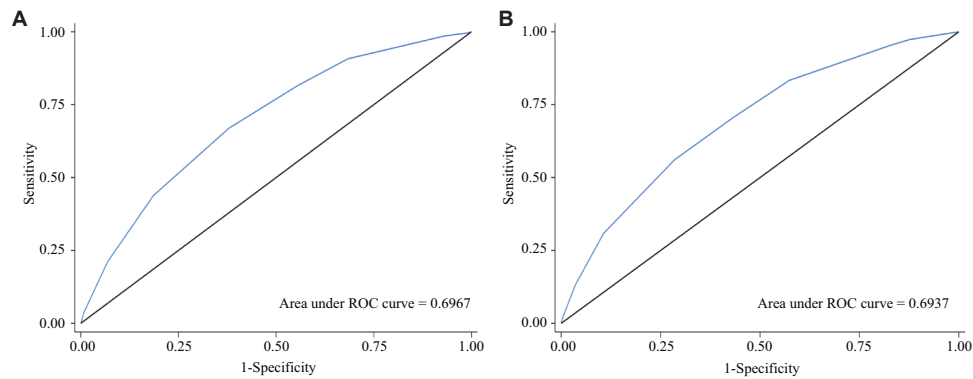


Fig. 1. ROC curve showing the performance of MDRF-IDRS among the (A) urban and (B) rural population in diagnosing diabetes. ROC, receiver operating characteristic; MDRF-IDRS, Madras Diabetes Research Foundation-Indian Diabetes Risk Score

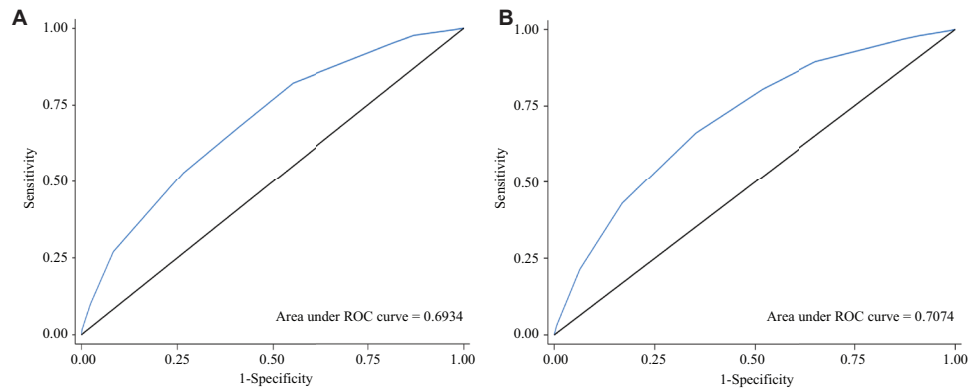


Fig. 2. ROC curve showing the performance of MDRF-IDRS among (A) male and (B) female population in diagnosing diabetes. ROC, receiver operating characteristic; MDRF-IDRS, Madras Diabetes Research Foundation-Indian Diabetes Risk Score

Table IV. Proportion of diabetic subjects identified by the MDRF-IDRS^a in the study population – by regions

MDRF-IDRS classification and test performance	North	South	East	West	Central	North East	Total
Newly diagnosed diabetes, n	1090	851	471	661	279	951	4303
Low risk, n (%)	33 (3.0)	19 (2.2)	30 (6.4)	24 (3.6)	22 (7.9)	39 (4.1)	167 (3.9)
Moderate risk, n (%)	342 (31.4)	264 (31.0)	201 (42.7)	230 (34.8)	114 (40.9)	396 (41.6)	1547 (35.9)
High risk, n (%)	715 (65.6)	568 (66.7)	240 (51.0)	407 (61.6)	143 (51.3)	516 (54.3)	2589 (60.2)
Combined moderate/high risk, n (%)	1057 (97.0)	832 (97.8)	441 (93.6)	637 (96.4)	257 (92.1)	912 (95.9)	4136 (96.1)
Sensitivity, %	65.60	66.75	50.96	61.57	66.97	54.26	60.17
Specificity, %	63.07	62.81	75.69	65.93	64.65	73.97	68.84
Positive predictive value, %	8.14	9.71	8.43	8.47	10.10	6.67	8.11
Negative predictive value, %	97.35	96.92	97.23	97.10	97.06	97.92	97.42
Accuracy, %	63.19	63.04	74.65	65.71	72.97	73.32	68.46

^aMDRF-IDRS risk score: <30 denotes that the risk of having diabetes is probably low; 30-50 denotes moderate risk and ≥60 denotes high risk of having diabetes. North: Chandigarh, Delhi, Haryana, Himachal Pradesh, Punjab, Uttarakhand and Uttar Pradesh; South: Andhra Pradesh (undivided), Karnataka, Kerala, Puducherry and Tamil Nadu; East: Bihar, Orissa, Jharkhand and West Bengal; West: Rajasthan, Gujarat, Goa and Maharashtra; Central, Madhya Pradesh and Chhattisgarh; North East: Assam, Sikkim, Nagaland, Meghalaya, Manipur, Mizoram, Tripura and Arunachal Pradesh

study. First, the cross-sectional nature of the study did not allow for inferences of causality to be made

and the validation of the score with incident diabetes was not performed. Although cross-sectional data are

adequate for detecting undiagnosed diabetes using risk score, prospective data would provide additional useful insight. Second, some of the differences could be attributed to the time lag in data collection between various phases of the study, which, however, is inevitable when sampling a country as big as India. However, as the objective of this current study was to evaluate the use of MDRF-IDRS in different parts of the country and the country as a whole, this time lag in the survey period does not affect the results. Our results also do not provide information on the prevalence of diabetes in individuals younger than 20 yr because this was beyond the scope of the study. Finally, our study methodology did not allow differentiating between type 2 diabetes and type 1 diabetes, latent autoimmune diabetes of adults (LADA) and maturity onset diabetes of the young (MODY) .

In conclusion, validation of MDRF-IDRS in Indian population has clinical implications. Through this study, the performance of MDRF-IDRS was evaluated across the nation. The study findings underscore the needs for using this validated score-tool in diabetes prevention intervention programmes. Such an approach may aid in an economic way of screening, conducted at the individual as well as the healthcare provider level. As the score can be self-assessed, it provides an inexpensive, easy and secure way to comprehend risk factors and empower the high-risk population for screening appropriate help. In addition, as the predictors included in the score keep changing with time and changes in lifestyle, periodic usage of the score in the same population will add value. Thus, identifying the high-risk group for diabetes will pave the way for effective intervention programmes, which is a desirable goal in public health policy and clinical practice.

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Supplementary Table I. Proportion of self-reported individuals with diabetes identified by the Madras Diabetes Research Foundation-Indian Diabetes Risk Score (MDRF-IDRS)^a in the study population – by regions

MDRF-IDRS classification and test performance	North	South	East	West	Central	North East	Total
Self-reported diabetes, n	1271	1606	541	951	270	1073	5712
Low risk, n (%)	11 (0.9)	11 (0.7)	12 (2.2)	9 (0.9)	5 (1.9)	15 (1.4)	63 (1.1)
Moderate risk, n (%)	221 (17.4)	328 (20.4)	160 (29.6)	126 (13.2)	71 (26.3)	310 (28.9)	1216 (21.3)
High risk, n (%)	1039 (81.7)	1267 (78.9)	369 (68.2)	816 (85.8)	194 (71.9)	748 (69.7)	4433 (77.6)
Combined moderate/high risk, n (%)	1260 (99.1)	1595 (99.3)	529 (97.8)	942 (99.1)	265 (98.1)	1058 (98.6)	5649 (98.9)
Sensitivity, %	81.75	78.89	68.21	85.80	83.54	69.71	77.61
Specificity, %	63.07	62.81	75.69	65.93	64.65	73.97	68.84
Positive predictive value, %	11.40	19.36	12.40	15.64	16.88	9.39	13.13
Negative predictive value, %	98.35	96.34	97.93	98.44	97.86	98.44	98.06
Accuracy, %	64.10	64.45	75.34	67.29	73.79	73.81	69.34

^aMDRF-IDRS risk score: <30 denotes that the risk of having diabetes is probably low; 30-50 denotes moderate risk and ≥60 denotes high risk of having diabetes. North: Chandigarh, Delhi, Haryana, Himachal Pradesh, Punjab, Uttarakhand and Uttar Pradesh; South: Andhra Pradesh (undivided), Karnataka, Kerala, Puducherry and Tamil Nadu; East: Bihar, Orissa, Jharkhand and West Bengal; West: Rajasthan, Gujarat, Goa and Maharashtra; Central: Madhya Pradesh and Chhattisgarh; North East: Assam, Sikkim, Nagaland, Meghalaya, Manipur, Mizoram, Tripura and Arunachal Pradesh

Supplementary Table II. Proportion of individuals with pre-diabetes identified by the MDRF-IDRS^a in the study population by regions

MDRF-IDRS classification and test performance	North	South	East	West	Central	North East	Total
Pre-diabetes, n	3733	2631	1586	2152	1436	3958	15496
Low risk, n (%)	266 (7.2)	130 (5.2)	154 (9.8)	146 (6.8)	215 (15.1)	416 (10.6)	1327 (8.7)
Moderate risk, n (%)	1646 (44.8)	982 (39.3)	777 (49.5)	898 (42)	715 (50.2)	1933 (49.3)	6951 (45.7)
High risk, n (%)	1761 (47.9)	1384 (55.4)	639 (40.7)	1093 (51.1)	494 (34.7)	1571 (40.1)	6942 (45.6)
Combined moderate/high risk, n (%)	3407 (92.8)	2366 (94.8)	1416 (90.2)	1991 (93.2)	1209 (84.9)	3504 (89.4)	13,893 (91.3)
Sensitivity, %	47.94	55.45	40.70	51.15	34.69	40.08	45.61
Specificity, %	65.29	66.71	78.51	69.31	76.17	76.28	71.63
Positive predictive value, %	21.81	26.22	24.51	24.84	28.08	21.77	23.66
Negative predictive value, %	86.13	87.53	88.53	87.74	81.30	88.54	87.23
Accuracy, %	61.05	62.95	71.97	64.95	66.65	70.32	66.22

^aMDRF-IDRS risk score: <30 denotes that the risk of having diabetes is probably low; 30-50 denotes moderate risk and ≥60 denotes high risk of having diabetes. North: Chandigarh, Delhi, Haryana, Himachal Pradesh, Punjab, Uttarakhand and Uttar Pradesh; South: Andhra Pradesh (undivided), Karnataka, Kerala, Puducherry and Tamil Nadu; East: Bihar, Orissa, Jharkhand and West Bengal; West: Rajasthan, Gujarat, Goa and Maharashtra; Central: Madhya Pradesh and Chhattisgarh; North East: Assam, Sikkim, Nagaland, Meghalaya, Manipur, Mizoram, Tripura and Arunachal Pradesh

States/Union Territories	Urban				Rural				Overall			
	n	Low	Moderate	High	n	Low	Moderate	High	n	Low	Moderate	High
Uttaranchal	49	1 (2)	11 (22.4)	37 (75.5)	70	1 (1.4)	29 (41.4)	40 (57.1)	119	2 (1.7)	40 (33.6)	77 (64.7)
West Bengal	67	2 (3)	18 (26.9)	47 (70.1)	105	4 (3.8)	41 (39.0)	60 (57.1)	172	6 (3.5)	59 (34.3)	107 (62.2)
Overall	1614	40 (2.5)	494 (30.6)	1080 (66.9)	2689	127 (4.7)	1053 (39.2)	1509 (56.1)	4303	167 (3.9)	1547 (36.0)	2589 (60.2)

Values are presented as number of subjects (%). ^aMDRF-IDRS risk score: <30 denotes that the risk of having diabetes is probably low; 30-50 denotes moderate risk and ≥60 denotes high risk of having diabetes

Supplementary Table IV. Validation of MDRF-IDRS in various studies

Author, yr	Place of study	Study population	Sample size	Findings
Adhikari <i>et al</i> ¹³ , 2010	Mangalore, coastal Karnataka	Bolloor locality	551	MDRF-IDRS ≥ 60 : Sensitivity - 62.2 %; Specificity - 73.7 %
Bhadoria <i>et al</i> ²¹ , 2015	Madhya Pradesh	Jabalpur district	911	MDRF-IDRS ≥ 40 : Sensitivity - 60.4 %; Specificity - 70.7 % MDRF-IDRS ≥ 60 : Sensitivity - 26.4 %; Specificity - 91.7 %
Nagarathna <i>et al</i> ¹⁶ , 2020	Nationwide diabetes screening	Nationwide diabetes screening	240,000	MDRF-IDRS ≥ 50 : Sensitivity - 78.1 %; Specificity - 63.0 % MDRF-IDRS ≥ 60 : Sensitivity - 61 %; Specificity - 77 %

Appendix. Sampling and weightage analysis – The Indian Council of Medical Research–India DIABetes study

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Sampling

Sample size calculation: The sample size was calculated separately for urban and rural areas (Table SI), as previous studies have shown large variations in urban and rural prevalence of type 2 diabetes mellitus. Assuming a prevalence of 10 per cent in urban areas and four per cent in rural areas, allowing for margin of error of 20 per cent, a non-response rate of 20 per cent and level of significance of five per cent, the sample size was estimated to be 1200 in urban areas and 2800 in rural areas in each of the states studied, with a total of 4000 individuals per state.

Sampling design: A stratified multi-stage sampling design [similar to the one employed in the National Family Health Survey - 3 (NFHS-3)] was adopted for this study (Fig. S1). A two-stage design (village-household) was used in rural areas, while a three-stage design [wards-census enumeration blocks (CEBs)-household] was adopted in urban areas. In both urban and rural areas, three-level stratification was done based on geographical distribution, population size and female literacy rate [as a surrogate of socio-economic status (SES)] so as to provide a sample of individuals that was truly representative of the population of the state under study.

The first level of stratification was based on geographic distribution with each state/union territory (UT) being divided into contiguous districts. This was based on the NFHS-3 sampling methodology. The second level of stratification was based on population size to ensure that there was no bias in the study and that all villages/wards, big and small, were represented in the sample studied proportionate to their contribution to the total rural/urban population of a state. The probability proportional to population size method was used to achieve this. The third level of stratification was based on the rural/urban female literacy rate, which was used as a surrogate of SES, to ensure that the sample of villages/wards selected was truly representative of the state studied. The primary sampling units (PSUs) were villages in rural areas and CEBs in urban areas. The ultimate stage units were households in both areas. Households were selected by systematic sampling with a random start. From each urban and rural PSU, 24 and 56 households were selected, respectively. In both rural and urban areas, only one individual was selected within each household using the World Health Organization (WHO) 'Kish method' (STEPwise approach to surveillance WHO, <http://www.who.int/chp/steps/en/>).

Example of sample selection in rural areas: In rural areas, the 2001 Census list of villages served as the sampling frame. To ensure that the villages included in the study were representative of the rural population of the state under study, three levels of stratification were used in each state. The first level of stratification was geographic, with the state being divided into contiguous regions. This stratification was adopted from the NFHS-3. Table SII shows the stratification of regions for Tamil Nadu state.

The second level of stratification was based on the village population size (Table SIII). This variable was included to ensure that there was no bias in the study and that all villages, big and small, were represented in the sample studied proportionate to their contribution to the total rural population. Arbitrary cut-offs were chosen as shown in Table SIII.

The third and final level of stratification was done based on female literacy rate (Table SIV). This variable was chosen as a crude predictor of the SES of the state. The female literacy rate for each state will be studied, and then, median cut-offs will be made to ensure equal and even stratification.

All the variables used for stratification have been used in NFHS-3 design as well. For the purpose of simplicity, we have chosen only one state (Tamil Nadu) to illustrate the details of the methodology used in the study.

After the first level of stratification, Tamil Nadu state was divided into five regions (as shown in Table SII). Table SV shows the rural population in each region of Tamil Nadu and the number of villages in each of these regions. Thus, in Tamil Nadu, the number of villages in each region was calculated as shown in Table SV.

Table SVI shows Stratification II and III done for Region 1 in rural Tamil Nadu for selecting the nine villages. Table SVII shows the list of villages selected in Region 1 along with their district codes. Fig. S2 shows the sampling framework in rural areas.

Example of sample selection in urban areas: For Stratification I, the Tamil Nadu state was divided into regions similar to the rural areas as shown in Table SII. In the next level of stratification, all urban towns in the given region were stratified accordingly to population size. Table SVIII shows the second level of stratification used for urban areas.

The third level of stratification was done using female literacy rate. Just as in the rural areas, the median of literacy was used for the stratification. Table SIX shows stratification III in urban areas.

Table SX shows the region-wise selection of wards in urban Tamil Nadu. Just as for villages in rural areas, it was decided to study 50 wards in each state.

Table SXI shows Stratification II and III done in urban Tamil Nadu, and Table SXII shows the list of wards that were selected in region 1. Fig. S3 shows the sampling framework in urban areas.

Estimation procedure adopted to arrive at the weights

Weights: Weighting was done for urban and rural areas separately for each state.

The development of weights for the Indian Council of Medical Research–India DIABetes (ICMR–INDIAB) study involved three steps:

Step 1 - The base weight (BW) or design weight or the ratio of probability of selection to probability of selection in the domain.

$$BW = \frac{\text{Probability of selection}}{\text{Probability of selection in the domain}}$$

Step 2 - An adjustment for non-response of individuals eligible for the survey. Non-response adjustment is done at the individual level. In each state/UT household level non-response is considered separately in urban and rural areas.

$$\text{Adjusted } BW (ABW) = \frac{BW}{\text{Response rate (RR)}}$$

where

$$RR = \frac{\text{Total number of responders}}{\text{Total sample size}}$$

Step 3 - After adjustment for non-response, the weights are normalized so that the total number of weighted cases is equal to the total number of unweighted cases. This is done by multiplying the final weight by the number of responders.

Normalization of weight is done using the following formulae:

$$\text{Normalized weight (NW)} = ABW \times \text{Number of responders}$$

Then, developing weights pooling across urban and rural:

$$\text{Weight (Urban)} = ABW(\text{Urban}) \times \frac{\text{Total number of responders}}{\text{Overall NW}}$$

$$\text{Weight (Rural)} = ABW(\text{Rural}) \times \frac{\text{Total number of responders}}{\text{Overall NW}}$$

Ultimately, the final weight (W) for an individual respondent was computed as the product of the NWs (separately in each geographical region within rural/urban area) and domain weight (urban and rural separately). The final weights were used in all analyses to produce estimates of population parameters. An example for weight calculation for the state of Tamil Nadu is provided in Tables SXIII-SXV.

Survey periods in The Indian Council of Medical Research–India DIABetes (ICMR–INDIAB) study: In Phase I, four regions representing the south (Tamil Nadu), north (Chandigarh), east (Jharkhand) and west (Maharashtra) of the country were studied from 2008 to 2010. Between 2011 and 2020, the remaining states were surveyed as follows: Phase II consisted of undivided Andhra Pradesh (subsequently divided into Andhra Pradesh and Telangana), Bihar, Gujarat, Karnataka and Punjab (survey period, 2012-2013), Phase III included Delhi, Madhya Pradesh, Rajasthan and Uttar Pradesh (survey period, 2017-2018), Phase IV included Kerala, Goa, Puducherry, Haryana and Chhattisgarh (survey period, 2018-2019), North East Phase included Assam, Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim and Tripura (survey period, 2011-2017) and Phase V included Himachal Pradesh, Uttarakhand, Odisha and West Bengal (survey period, 2019-2020).

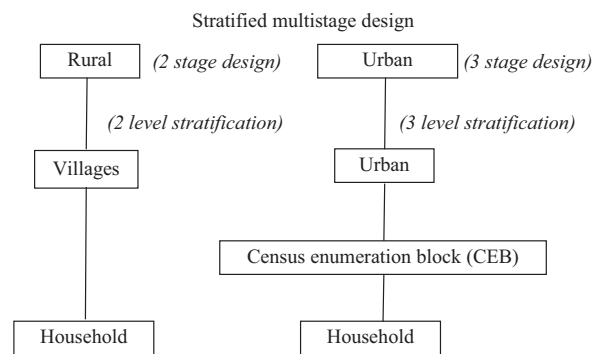


Fig. S1. The Indian Council of Medical Research–India DIABetes sampling strategy. Source²⁶. ICMR–INDIAB, Indian Council of Medical Research–India DIABetes.

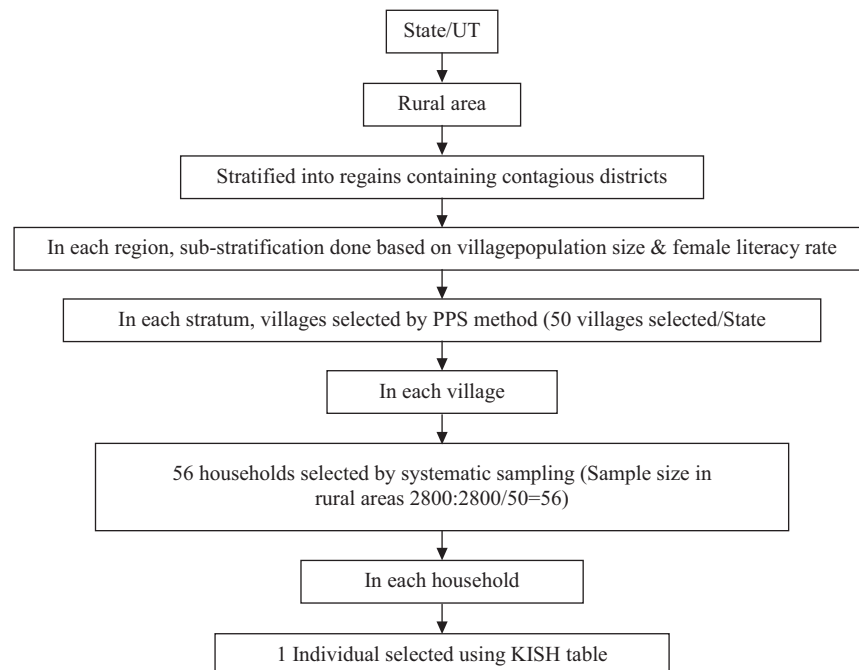


Fig. S2. The Indian Council of Medical Research–India DIABetes sampling framework in rural areas. Source²⁶. ICMR–INDIAB, Indian Council of Medical Research–India DIABetes; UT, union territory; PPS, probability proportional to population size.

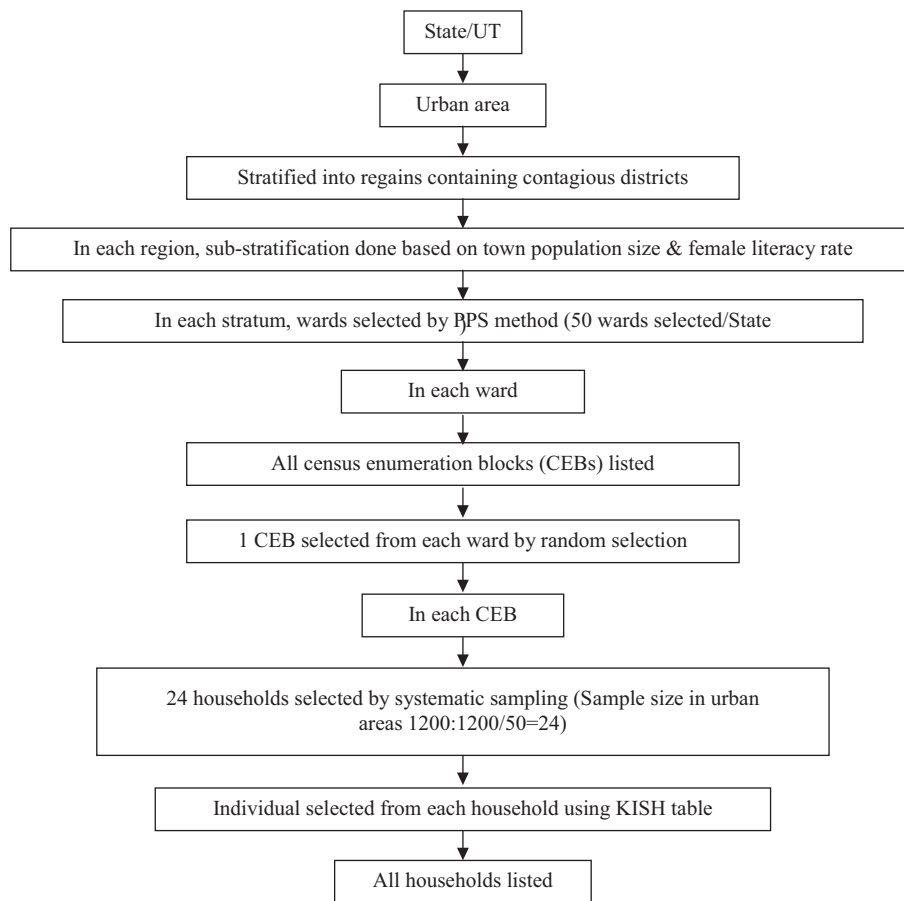


Fig. S3. The Indian Council of Medical Research–India DIABetes sampling framework in urban areas. Source²⁶. ICMR–INDIAB, Indian Council of Medical Research–India DIABetes; UT, union territory.

Table SI. Sample size calculation for phase II of the Indian Council of Medical Research-INDia DIABetes study					
Sample size calculation ICMR-INDIAB	Prevalence (p)	q=(1-p)	Margin of error (d)=20 per cent of P	Sample size per state (n)	Rounded off to
				$n = \frac{Z^2 (p)(q)}{d^2}$	
Rural	Diabetes=0.04	0.96	0.008	$n = \frac{(1.96)^2 (0.04)(0.96)}{(0.008)^2} = 2304.96$ $n = \frac{20 \times 2305}{100} = 461$	2800
Urban	Diabetes=0.10	0.90	0.02	2305 $n = \frac{(1.96)^2 (0.10)(0.90)}{(0.02)^2} = 864.36$ $n = \frac{20 \times 865}{100} = 173$ $865+173=1038$	1200
Formula: Sample size $n = \frac{Z^2 (p)(q)}{d^2}$ Z=Z statistic for a level of confidence. For the level of confidence of 95 per cent, the conventional Z value is 1.96 {2 SD}; P=Prevalence or proportion of the aspect being studied in the population; q=(1-p); d=Margin of error of the estimated prevalence					

Table SII. Stratification I – Regions and districts in Tamil Nadu (based on the National Family Health Survey-3 methodology)

Regions	Districts
1	Coimbatore, Dindigul, Madurai, Theni, Erode, Nilgiri
2	Vellore, Dharmapuri, Krishnagiri, Tiruvannamalai, Salem, Namakkal, Tiruchirappalli, Karur, Perambalur, Ariyalur
3	Kanyakumari
4	Kanchipuram, Tiruvallur, Cuddalore, Villupuram, Thanjavur, Nagapattinam, Tiruvarur, Chennai
5	Pudukkottai, Sivaganga, Virudhunagar, Ramanathapuram, Thoothukudi, Tirunelveli

NFHS, National Family Health Survey

Table SIII. Stratification II - Based on the village population size

Stratification II (village population size)

<1000

1000-3000

>3000

Table SIV. Stratification III - Based on the female literacy rate

Stratification III (female literacy)

Below median

Above median

Table SV. Region-wise selection of number of villages in the state

Region	Rural population (a)	Number of villages	Villages selected for the present study [#]
1	6,032,618	1844	9
2	11,864,028	4775	17
3	582,107	69	3
4	10,220,031	5705	15
5	6,222,897	3007	9
Total	34,921,681 (b)	15,400	53

Number of villages to be selected for the study=100 per state. $\frac{a}{b} \times \text{total number of villages to be selected per state}$ (*i.e.* n=100)

Table SVI. Stratification II and III for region 1 in rural Tamil Nadu					
Female literacy rate	Population	Number of villages	Per cent total population	Per cent of villages	Number of villages to be selected
Population size <1000					
<46.95	108,149	206	1.79	0.16	1
≥46.95	86,183	161	1.43	0.13	1
Population size 1000-3000					
<46.95	876,122	461	14.52	1.30	1
≥46.95	482,504	256	8.0	0.72	1
Population size >3000					
<46.95	2,031,918	382	33.68	2.03	2
≥46.95	2,447,742	378	40.58	3.45	3
Total					9

Table SVII. Number of villages selected for the study (Region 1)							
Population size (n)	Female literacy (%)	Total population	Number of villages	Number of villages selected	Village name	District code	Sub-district code
<1000	<46.95	108,149	206	1	Sirupattii	24	0004
	≥46.95	86,183	161	1	Idayanathan	24	0006
1000-3000	<46.95	876,122	461	1	Sengattampatti	13	0007
	≥46.95	482,504	256	1	Sellappampalayam	12	0008
>3000	<46.95	2,031,918	382	2	Kadamalaikundu	25	0005
					Thullukkuttinayakkanur	24	0005
	≥46.95	2,447,742	378	3	Cherangode	11	0001
					T. Meenakshipuram	25	0004
					Kandiankoil	12	0003

Table SVIII. Stratification II - Based on the town population size
Stratification III (town population size)
<20,000
20,000-100,000
>100,000

Table SIX. Stratification III - Based on the Female literacy rate
Stratification III (female literacy)
Below median
Above median

Table SX. Region wise selection of number of wards in Tamil Nadu state

Region	Urban population (a)	Number of wards	Wards selected for the present study [#]
1	7,178,044	4095	13
2	5,708,744	3482	11
3	1,093,927	1029	2
4	9,876,057	3295	18
5	3,627,226	2382	8
Total	27,483,998 (b)	14,283	52

Number of wards to be selected for the study=50 per state. $\frac{a}{b} \times \text{Total number of wards to be selected per state}$ (*i.e.* n=50)

Table SXI. Stratification II and III for region I in urban Tamil Nadu

Female literacy rate	Population	Number of towns	Number of wards	Per cent of wards	Number of wards to be selected
Population size <20,000					
<65.31	16,937	30	0.24	0.03	1
≥65.31	0	0	0	0	0
Population size 20,000-100,000					
<65.31	782,355	656	10.90	1.42	1
≥65.31	160,543	126	2.24	0.29	1
Population size >100,000					
<65.31	2,659,331	1991	37.05	4.41	4
≥65.31	3,558,878	1292	49.58	6.44	6
Total					13

Total number of wards selected in region I=13

Table SXII. List of wards selected for the study (Region 1)

Population size (n)	Female literacy (%)	Total population	Number of towns	Number of wards selected	Town (ward number)	District code	Sub-district code
<20,000	<65.31	16937	30	1	Andipalayam (CT) - Ward number 2	12	0003
	≥65.31	0	0	1	-	12	0005
20,000-100,000	<65.31	782,355	656	1	O' Valley (TP) - Ward number 7	11	0002
	≥65.31	160,543	126	4	Theni Allinagaram (M) - Ward number 27	25	0003
					Theni Allinagaram (M) - Ward number 23	25	0003
					Kotagiri (TP) - Ward number 7	11	0004
					Periyanaickenpalayam (TP) - Ward number 13	12	0005
>100,000	<65.31	2,659,331	1991	0	-	-	-
	≥65.31	3,558,878	1292	6	Coimbatore (M.Corp.) - Ward number 54	12	0006
					Coimbatore (M.Corp.) - Ward number 19	12	0006
					Coimbatore (M.Corp.) - Ward number 53	12	0006
					Madurai (M.Corp.) - Ward number 25	24	0007
					Coimbatore (M.Corp.) - Ward number 32	12	0006
					Erode (M) - Ward number 19	10	0005

Table SXIII. Example of weight calculation used for Tamil Nadu state (as an example)

Tamil Nadu State (A)	Details (B)	Urban (C)	Rural (D)	Total (E)	Cell number 1
					2
Population	Total population	27,483,998	34,921,681	62,405,679	3
Sample size	Total sample size	1248	2968	4,216	4
Probability of selection (fi)	C4/C3 (for urban) D4/D3 (for rural)	0.00005	0.00008	0.0000676	5
Number screened		1076	2588	3664	6
RR	C6/C4 (for urban) D6/D4 (for rural)	0.86217	0.87196	0.86907	7
Design weight (ratio of probability of selection and probability of selection in the domain)	E5/C5 (for urban) E5/D5 (for rural)	1.48779	0.79489		8
Weight adjusted for RR (design weight/RR)	C8/C7 (for urban) D8/D7 (for rural)	1.7256	0.91		9
Normalization of weight	C9×C6 (for urban) D9×D6 (for rural)	1856.76268	2359.23732	4216	10
Final weight	C9×(E4/C10+D10) (for urban) D9×(E4/D10+C10) (for rural)	1.49968	0.79224	0.86907	11
Weighted number of cases	C6×C11 (for urban) D6×D11 (for rural)	1613.65713	2050.34287	3664	12
RR, response rate					

Table SXIV. Example of weight calculation used for Tamil Nadu state - rural (as an example)

Rural Tamil Nadu	Region 1	Region 2	Region 3	Region 4	Region 5	Total
Population	6,032,618	11,864,028	582,107	10,220,031	6,222,897	34,921,681
Sample	504	952	168	784	560	2968
Actual tests	448	824	149	716	451	2588
RR	0.88888	0.86554	0.88690	0.91326	0.80535	0.87196
Probability of selection	0.000083	0.000080	0.00028	0.000076	0.000089	0.000084
Design weight	1.01728	1.05916	0.29448	1.10791	0.94443	
Weight adjusted for RR	1.14444	1.22369	0.33203	1.21313	1.17269	
Number of cases	512.71	1008.33	49.47	868.60	528.89	2968.00
Normalized weight	0.99792	1.06702	0.28952	1.05781	1.02255	
Number of cases	447.06941	879.22756	43.13918	757.39309	461.17073	2588.00
Weights for overall rural	0.79224	0.79224	0.79224	0.79224	0.79224	
Final weights	0.79060	0.84534	0.22937	0.83805	0.81011	
RR, response rate						

Table SXV. Example of weight calculation used for Tamil Nadu state – Urban (as an example)

[illegible]